Renormalization Obstructions in Congruence and Homology Towers of Geodesic Flows

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Abstract

This comparative methods and calibration note separates periodic owners in residual inverse limits from finite-panel lift factors in a different homology-cover candidate. For every descending normal finite-index tower with trivial intersection, every infinite-order owner's quotient orders divide forward and tend to infinity; with one common physical clock, the coordinatewise inverse-limit geodesic flow has no periodic points, and each fixed-owner Euler factor becomes 1 on every fixed coefficient prefix. We specialize this prior residual-solenoid aperiodicity mechanism to the cusped chain $\Gamma(3n!)$ with an explicit projective-sign proof and compare it with a closed genus-two residual/homology control, where primitive homology gives factorial lower bounds for minimal lifted periods. We then audit the distinct pure homology-cover calibrator $H_N = \ker(\Gamma \to H_1(\Sigma; \mathbb{Z}/N\mathbb{Z}))$. For a primitive content-one owner, its cover degree, deck order, primitive lift count, and lift period are N^4, N, N^3, and $N\ell(g)$. For every N and each fixed finite owner panel, the four registered clock/multiplicity factors are exact in the coefficientwise formal-series topology: physical time gives support escape, $1/N$ time rescaling alone gives multiplicity divergence, and $1/N^3$ logarithmic normalization alone leaves support escape; among these four quadrants, only their combination recovers $(1-x_g)^{-1}$. The recovery is target-matched rather than canonical, changes owner declaration, tower, clock, normalization, and finite-panel scope, and transfers no Route credit to the residual candidate. Neither candidate supplies an arithmetic dynamical determinant or advances beyond its unchanged rejected Route-A tuple; Route B remains unavailable.

Keywords: inverse-limit flow; residual tower; closed geodesic; quotient-order escape; homology cover; Euler-factor renormalization

繁體中文摘要

本文是一篇比較方法與校準註記，嚴格區分殘餘逆極限中的週期所有者，以及另一個同調覆蓋候選中的有限面板提升因子。對任一交集為單位元的下降正規有限指數塔，每個無限階所有者在有限商中的階數向前整除且趨於無限；若各層共用同一物理時鐘，座標式逆極限測地流沒有週期點，且每個固定所有者的 Euler 因子在每個固定係數前綴上最終等於 1。本文把此既有的殘餘 solenoid 非週期機制特化到尖點鏈 $\Gamma(3n!)$，給出明確的投影正負號證明，並與封閉 genus-two 殘餘／同調控制比較；原始同調給出最小提升週期的階乘下界。其後另行稽核純同調覆蓋校準器 H_N。對內容為一的原始所有者，覆蓋次數、deck 階、原始提升分量數與週期分別為 N^4、N、N^3 與 $N\ell(g)$。對每個 N 與每個固定有限所有者面板，四個已註冊的時鐘／重數因子在係數逐項的形式冪級數拓撲中皆為精確恆等式：物理時間造成支撐逃逸；僅用 $1/N$ 時間縮放造成重數發散；僅用 $1/N^3$ 對數正規化仍有支撐逃逸；在這四個象限之內，只有兩者合用才恢復 $(1-x_g)^{-1}$。此恢復是目標匹配而非典範構造，並同時改變所有者宣告、覆蓋塔、時鐘、正規化與有限面板範圍，因此不向殘餘候選轉移任何 Route credit。兩個候選均未給出算術動力行列式，其遭拒的 Route-A tuple 均保持不變；Route B 仍不可用。

關鍵詞： 逆極限流；殘餘塔；閉測地線；商階逃逸；同調覆蓋；Euler 因子重整化

1 Introduction

Periodic-orbit constructions require an unambiguous owner. This requirement becomes nontrivial in a tower of finite covers. Each finite cover can have closed geodesics, and a fixed base geodesic can close after a level-dependent number of traversals. A periodic point in the inverse limit, however, requires one coherent point and one common positive time that returns at every coordinate. Confusing these two statements produces an apparent orbit population for a limit flow that may in fact have none.

This paper studies two related questions. The first is an owner-preserving no-go problem: what happens to the periodic orbits and finite Euler factors of a fixed base owner in a normal residual tower with the same physical clock at every level? The second is a bounded calibration problem: after owner declaration, cover tower, clock, and multiplicity normalization are explicitly changed, which of the four registered clock/exponent combinations recovers a fixed finite panel coefficientwise? No claim is made about a growing panel or a global analytic determinant.

The first answer is rigid. In a descending normal finite-index tower with trivial intersection, the finite quotient order of every infinite-order element tends to infinity. A coherent inverse-limit period would put one fixed nonzero power into every subgroup and is therefore impossible. In physical time, the lifted period of every fixed owner escapes every bounded interval. Algebraically, the corresponding factor $(1 - x_g^{o_n(g)})^{-1}$ converges coefficientwise to 1 on every fixed prefix.

The second answer is exact within the registered four-quadrant scalar clock/exponent comparison. In a pure genus-two homology cover of modulus N, a primitive content-one owner has deck order N, exactly N^3 primitive lifts, and physical period $N\ell(g)$. Keeping physical time gives support at degree N; applying the target-matched $1/N$ time rescaling repairs that support but leaves multiplicity N^3. Applying the target-matched $1/N^3$ logarithmic normalization repairs that multiplicity but, without time rescaling, leaves support at degree N. Among these four interventions, their combination returns the base factor exactly for every N in the coefficientwise topology of each fixed finite panel. This bounded statement does not assert uniqueness outside the scalar clock/exponent class.

The article contributes a unified proof chain rather than a broad priority claim. Aperiodic laminated geodesic flows, universal and punctured solenoids, and finite-type hyperbolic solenoidal surfaces already have direct literature. Our contribution is the explicit $\Gamma(3n!)$ specialization with its projective-sign residual proof, a compact-versus-cusped control, a coefficientwise owner firewall, and the four-quadrant theorem for a separately registered finite-panel calibrator. The distinction between those two candidates is maintained throughout.

Table 1: Non-ranking candidate identity and Route ledger.

Candidate	owner and tower	clock, normalization, panel, and status
Residual inverse limit	A fixed infinite-order base owner in a descending normal residual tower, with the explicit cusped specialization $\Gamma(3n!)$	One common physical arclength clock; no multiplicity renormalization; same-owner statements on each fixed finite panel; residual tuple below; Route A rejected.
Homology calibrator	A separately declared primitive content-one base-owner panel and its lift-component families in the nonresidual tower H_N	Physical period $N\ell(g)$ followed, in Q_{11}, by target-matched $1/N$ clock and $1/N^3$ logarithmic normalization; fixed finite panel only; calibrator tuple below; Route A rejected.

The exact non-ranking statuses are

$$\mathcal{C}_{\mathrm{res}}: \quad \begin{pmatrix} \text{A0\_WEAK\_ARITHMETIC\_RELATION, A1\_FAIL,} \\ \text{A2\_FAIL, A3\_FAIL, A4\_FAIL} \end{pmatrix}, \\ \texttt{ROUTE\_A\_REJECTED},$$

$$\mathcal{C}_{\mathrm{hom}}: \quad \begin{pmatrix} \text{A0\_FAIL, A1\_PASS\_ANALYTIC,} \\ \text{A2\_FAIL, A3\_FAIL, A4\_FAIL} \end{pmatrix}, \\ \texttt{ROUTE\_A\_REJECTED}.$$

Here A0 is the intrinsic arithmetic-relevance gate, A1 the primitive-owner and periodic-orbit layer, A2 the stable dynamical-zeta or Fredholm-determinant layer, A3 the global analytic and arithmetic-operator layer, and A4 the natural unitary, scattering, or Hamiltonian liftability layer. Within these tokens, `WEAK_ARITHMETIC_RELATION` records only limited arithmetic provenance, `PASS_ANALYTIC` records an analytic proof within the stated candidate scope, and `FAIL` records absence of the layer required for Route promotion. `ROUTE_A_REJECTED` is a scientific classification, not a ranking or work order. The Q_{11} row changes the owner declaration, tower, clock, normalization, and fixed-finite-panel scope; it transfers no credit to $\mathcal{C}_{\mathrm{res}}$. Route B is unavailable for both candidates.

Both Route-A conclusions remain negative. The original residual inverse-limit flow has no primitive periodic-orbit population and therefore fails A1. The homology-cover identity supports A1 analytically only for its separately declared lift-component object: for every N it is coefficientwise exact on each fixed finite panel of primitive content-one owners, and the same cover calculation applies to every marked genus-two hyperbolic metric in the stated class. That metric-wide validity is a proves-too-much calibration, not arithmetic specificity. The calibrator has no rational-prime owner map, full primitive spectrum, global dynamical determinant, or natural operator lift. No prime or Riemann-zero table is used, and neither tuple advances to Route B.

2 Prior work and bounded positioning

Laminated geodesic and horocycle flows are defined leafwise. Martínez, Matsumoto, and Verjovsky give the standard leafwise flow framework, an explicit compact hyperbolic-lamination example without periodic geodesic-flow orbits, and a universal-solenoid example with simply connected leaves. [3, Section 2.2 and Examples 4, 6] These results rule out any claim that general laminated aperiodicity or the simply-connected-leaf mechanism is new here.

Penner and Šarić define the punctured solenoid through finite modular covers and describe its disk leaves. [5, Introduction and Definition 2.1] The directed system in that work differs from the single factorial principal-congruence chain below, but it is direct prior for the noncompact modular-solenoid setting.

Alcalde Cuesta and collaborators define the leafwise geodesic flow and the finite-type hyperbolic solenoidal object class, including inverse limits of regular covers under the McCord terminology. [1, Definitions 4, 5, and 7] Their study concerns horocycle dynamics and does not state the explicit factorial-chain propositions proved below.

For compact weak solenoids, the intersection of a group chain is naturally related to the fundamental group of the corresponding leaf. [2, Definition 5.5] We use this only as structural comparison: the noncompact congruence case receives a direct group proof. Residual finiteness of the closed surface group follows from the standard theorem that finitely generated linear groups are residually finite. [4, p. 1]

Accordingly, the general residual-solenoid and laminated-flow aperiodicity mechanism is prior. This paper is framed as a comparative methods and calibration note: its local contributions are the exact $\Gamma(3n!)$ specialization and projective-sign proof, the compact-versus-cusped owner audit, coefficientwise same-owner bookkeeping, and the four-quadrant audit of a separately registered target-matched calibrator.

The bounded literature search found no treatment of the literal chain $\Gamma(3n!)$ with this exact combination of specialization and audit surfaces, but that search statement is not proof of theorem priority.

3 Normal residual towers and the owner firewall

Let Γ be a torsion-free Fuchsian group and

$$\Gamma = \Gamma_1 \supseteq \Gamma_2 \supseteq \cdots$$

a descending sequence of normal finite-index subgroups with $\bigcap_n \Gamma_n = \{e\}$. Put $Y_n = \Gamma_n \backslash \mathbb{H}$ and let

$$M_\infty = \varprojlim_n T^1 Y_n$$

under the tangent maps of the finite coverings. Every coordinate carries the unit-speed geodesic flow, using one hyperbolic-arclength clock.

In weak-solenoid or lamination language, M_∞ is viewed leafwise: a leaf follows the compatible covering directions, while the inverse limit of the finite covering fibers supplies a totally disconnected transversal and the deck actions supply its holonomy. The coordinatewise geodesic flow moves along a leaf by the same hyperbolic-arclength time at every coordinate. Consequently, periodicity means that one coherent point returns after one common $T > 0$; a family of level-dependent closing times is not a period of the inverse-limit flow. This model is used only to orient the owner and clock quantifiers. No groupoid trace, invariant state, transfer operator, or Fredholm determinant is constructed or used.

Definition 3.1 (period owner)**.** A periodic point of M_∞ is a coherent point $x = (x_n)$ and a single $T > 0$ such that $g^T x_n = x_n$ at every level. A sequence of level-dependent closing times T_n is finite-tower data and is not a period of x.

For $g \in \Gamma$, define

$$o_n(g) = \mathrm{ord}(g\Gamma_n \text{ in } \Gamma/\Gamma_n).$$

Theorem 3.2 (residual-tower escape)**.** *The coordinatewise geodesic flow on M_∞ has no periodic point. For every infinite-order $g \in \Gamma$,*

$$o_n(g) \mid o_{n+1}(g), \qquad o_n(g) \longrightarrow \infty.$$

Proof. Suppose $x = (\Gamma_n h_n)$ has period $T > 0$. Compatibility with level one gives $h_n = \eta_n h_1$ for some $\eta_n \in \Gamma$. The level-one orbit is a closed geodesic, so there is a primitive hyperbolic $\gamma \in \Gamma$ and an integer $m \geq 1$ such that $T = m\ell(\gamma)$ and

$$h_1 a_T h_1^{-1} = \gamma^m.$$

Return at level n is equivalent to

$$\eta_n \gamma^m \eta_n^{-1} \in \Gamma_n.$$

Normality removes η_n, so $\gamma^m \in \Gamma_n$ for every n. Trivial intersection gives $\gamma^m = e$, contradicting hyperbolicity.

The quotient map $\Gamma/\Gamma_{n+1} \to \Gamma/\Gamma_n$ implies $o_n(g) \mid o_{n+1}(g)$. If this divisibility sequence were bounded, it would eventually stabilize at some r. Then $g^r \in \Gamma_n$ for every n, so $g^r = e$, contradicting the infinite order of g. □

If g is primitive, its axis stabilizer in a torsion-free Fuchsian group is $\langle g \rangle$. Hence

$$\Gamma_n \cap \langle g \rangle = \langle g^{o_n(g)} \rangle,$$

and the minimal period of each lifted component is $o_n(g)\ell(g)$. Without a primitivity certificate, $o_n(g)\ell(g)$ is only the minimal closing time among whole traversals of the chosen g loop. This terminology prevents finite rows from acquiring stronger orbit credit than their certificates support.

3.1 Quantifiers, coherence, and the role of normality

Theorem 3.2 uses two quantifier changes that are easy to miss. At a fixed level n, a base owner g closes after $o_n(g)$ traversals, so there are many finite-level periodic orbits. A point of the inverse limit is periodic only if there is one time T that closes every coordinate simultaneously. The sequence $o_n(g)\ell(g)$ supplies the former statement and, because it diverges, obstructs the latter. Choosing T after choosing n does not establish a period of the limit flow.

Coherence of the point also matters. A coordinate in the cover may differ from the level-one representative by a conjugator η_n. Return therefore initially places $\eta_n\gamma^m\eta_n^{-1}$, rather than γ^m, in Γ_n. Normality is exactly what removes this level-dependent conjugator. For a nonnormal chain, trivial intersection of the subgroups alone does not justify the same step; one would need control of the conjugates or of an appropriate normal core. The theorem is consequently stated for normal towers and is not silently extended to every inverse system of finite covers.

Residuality has a separate function. It turns membership of one fixed nontrivial power in every level into a contradiction, and it forces the divisibility sequence of quotient orders to be unbounded. Neither finite index nor nestedness alone has this consequence. For example, a chain with nontrivial intersection can retain a closed owner represented by an element of that intersection. Thus the no-period conclusion belongs to the package "normal, descending, finite index, trivial intersection, common clock." The later homology calibrator deliberately leaves this package and must be treated as a different candidate.

4 The factorial principal-congruence specialization

For $n \geq 1$, let

$$\Gamma_n = \Gamma(3n!) < \mathrm{PSL}_2(\mathbb{Z}).$$

The chain is nested, normal, and finite index. Its residual intersection requires care because principal congruence in the projective group allows a sign.

Lemma 4.1 (projective-sign intersection)**.**

$$\bigcap_{n\geq 1} \Gamma(3n!) = \{1\} \quad \textit{in } \mathrm{PSL}_2(\mathbb{Z}).$$

Proof. Let $[A]$ lie in every subgroup and choose $A \in \mathrm{SL}_2(\mathbb{Z})$. For each n there is $\epsilon_n \in \{\pm 1\}$ with

$$A \equiv \epsilon_n I \pmod{3n!}.$$

Reducing the level-$n+1$ congruence modulo $3n!$ shows $\epsilon_{n+1} = \epsilon_n$, since $3n!$ cannot divide 2. Thus one sign ϵ works at every level. Every entry of $A - \epsilon I$ is divisible by the unbounded sequence $3n!$, hence $A = \epsilon I$ and $[A] = 1$. □

Theorem 3.2 now gives $\mathrm{Per}(M_\infty) = \varnothing$ for this explicit cusped tower and quotient-order divergence for every hyperbolic owner.

The finite diagnostic freezes three matrices in $\Gamma(3)$ and eight moduli

$$3, 6, 18, 72, 360, 2160, 15120, 120960.$$

Their projective order sequences are

$$\begin{aligned} \texttt{G3-A}&: \quad 1, 3, 3, 6, 6, 36, 72, 288, \\ \texttt{G3-B}&: \quad 1, 1, 3, 12, 60, 360, 360, 2880, \\ \texttt{G3-C}&: \quad 1, 2, 6, 12, 12, 72, 72, 576. \end{aligned}$$

Sequential matrix multiplication and an independent finite-group-bound factor reduction agree in all 24 rows, and all 21 noninitial bonding transitions pass. These exact-integer rows illustrate the theorem, including plateaus, but do not prove asymptotic divergence by extrapolation. Positive-word primitivity is checked; full $\Gamma(3)$ conjugacy primitivity for these three loop rows remains unestablished and is not needed for quotient orders.

The plateaus are informative. Divisibility permits $o_{n+1}(g) = o_n(g)$ for several successive levels, so monotone growth should not be read as strict growth. What the proof guarantees is eventual escape from every fixed bound. The finite table tests the projective-sign conventions, quotient arithmetic, and bonding maps on the registered rows; the asymptotic conclusion comes instead from Lemma 4.1 and Theorem 3.2. This separation prevents a short factorial schedule from being presented as empirical evidence for an infinite limit.

A direct Stage-4 kernel fixture now exercises the literal matrix $-I$ modulo each of the eight registered moduli $3, 6, 18, 72, 360, 2160, 15120, 120960$. At every modulus, both the sequential-order and group-bound routines return projective order one with terminal scalar sign -1. This fixture is not a new registered owner. All 24 owner rows in the canonical diagnostic terminate with sign $+1$: every registered modulus is divisible by 3 and every registered owner lies in $\Gamma(3)$, so an owner power congruent to I modulo 3 cannot simultaneously be congruent to $-I$ modulo the registered modulus. The canonical owner directory is unchanged.

The two order algorithms provide a useful internal adversary. Direct multiplication can certify the first return but may be expensive, while factor reduction begins with a finite group-order bound and removes prime factors whenever the corresponding power is already projectively scalar. Their agreement checks both minimality and the sign quotient. It does not certify geodesic primitivity, which is a different owner question and is intentionally reported at a weaker level for these cusped diagnostics.

The two strategies are not independent arithmetic implementations. They use different search procedures, but both call the same `scalar_sign` routine and both ultimately use the same matrix-multiplication primitive. The direct $-I$ regression therefore localizes the previously unexercised sign branch in the shared kernel; agreement on that fixture must not be counted as replication by an independently implemented sign or multiplication kernel.

5 A cocompact residual control

Let Σ be a marked closed hyperbolic surface of genus two and $\Gamma = \pi_1(\Sigma)$. For $n \geq 1$, define R_n as the intersection of all normal subgroups of Γ of index at most n, let

$$H_n = \ker\big(\Gamma \to H_1(\Sigma; \mathbb{Z}/n!\mathbb{Z})\big), \qquad \Gamma_n = R_n \cap H_n.$$

Proposition 5.1 (closed residual/homology tower)**.** *The groups Γ_n form a descending normal finite-index residual tower. If the integral homology vector of g has content d, then*

$$\frac{n!}{\gcd(n!, d)} \mid o_n(g).$$

For content-one g, $n! \mid o_n(g)$. If g is represented by a closed geodesic and has primitive homology, that geodesic is primitive and its minimal lifted period is

$$T_n(g) = o_n(g)\ell(g) \geq n!\ell(g).$$

Proof. A finitely generated group has only finitely many subgroups of any fixed index, so R_n has finite index. Normality is built into the definition. Residual finiteness gives $\bigcap_n R_n = \{e\}$, and intersecting with H_n preserves trivial intersection. The factorial moduli make both chains descending.

The image of g in integral homology is a vector v of content d. Its additive order modulo $n!$ is $n!/\gcd(n!, d)$. Since the quotient by Γ_n maps onto the homology quotient, the order of $g\Gamma_n$ is divisible by this number. If v is primitive and $g = h^q$ with $q > 1$, then $v = q[h]$ is not primitive, a contradiction. Thus g is not a proper power. The cyclic axis stabilizer then identifies $o_n(g)\ell(g)$ with the exact minimal period of a lifted component. $\square$

The executable retains three content-one owners at $n = 1, \dots, 8$ and certifies the lower-bound sequence

$$1, 2, 6, 24, 120, 720, 5040, 40320.$$

It does not enumerate R_n or claim the full quotient orders were computed. This control removes cusps, principal congruence, and arithmetic-lattice provenance. The causal mechanism behind period escape is normal residuality with one common clock, not arithmetic specificity.

6 Coefficientwise same-owner escape

Let g be a fixed primitive owner and introduce the formal owner variable

$$x_g = e^{-s\ell(g)}.$$

At level n, its reciprocal unweighted factor under the physical clock is

$$E_{n,g}(x_g) = \left(1 - x_g^{o_n(g)}\right)^{-1}.$$

Theorem 6.1 (fixed-prefix escape)**.** *For every fixed integer $D \geq 0$, there is $n_0(g, D)$ such that*

$$E_{n,g}(x_g) = 1 \pmod{x_g^{D+1}}$$

for every $n \geq n_0(g, D)$. The same conclusion holds for any fixed finite panel of primitive owners. Finite level-dependent multiplicities and scalar weights cannot create a coefficient below degree $o_n(g)$.

Proof. The support of the geometric series is

$$0, o_n(g), 2o_n(g), \dots.$$

Theorem 3.2 gives $o_n(g) \to \infty$, so eventually $o_n(g) > D$. A finite exponent or scalar changes only the coefficients on the same support. For a finite panel, take the maximum of the finitely many thresholds. Equivalently, the minimum lifted physical period in the panel tends to infinity. □

This theorem strengthens the owner firewall. Not only are finite-level geodesics different from inverse-limit periodic points, but each fixed owner's formal Euler contribution becomes trivial on every fixed prefix if the original clock and owner identity are retained. It does not rule out a new collective candidate with a declared rescaled clock and normalization. Such a candidate must be registered separately, which is done next.

6.1 What fixed-prefix convergence does and does not say

The natural setting of Theorem 6.1 is the formal power-series ring in an owner variable. Fixed-prefix convergence means that for each D, the coefficients of degrees at most D eventually equal those of the constant series 1. It is stronger than observing that a factor is numerically close to one at a few chosen values of s, because every bounded formal degree is controlled at once. It is weaker than a global analytic convergence theorem: the theorem supplies no common domain, no bound uniform over an infinite owner family, and no justification for exchanging an inverse-limit operation with an infinite Euler product.

The variable x_g deliberately preserves owner identity. If several owners were prematurely projected to one scalar variable, equal or commensurable lengths could merge coefficients and obscure which factor produced them. The owner-indexed statement avoids that cancellation. After the same-owner result is proved, a finite panel may be specialized to numerical lengths because a maximum of finitely many escape thresholds still exists. A panel growing with n is different: newly introduced owners need not satisfy a uniform threshold, and the theorem makes no assertion about them.

Multiplicities also require precise wording. A finite exponent m_n at level n changes coefficients beginning at degree $o_n(g)$ but cannot create a lower-degree term. Hence arbitrary finite level-dependent lift counts leave fixed-prefix escape intact. Analytic norms can behave differently if m_n grows rapidly, since a large exponent can offset the smallness of $x_g^{o_n(g)}$ away from the formal origin. The four-quadrant analysis below is therefore conducted coefficientwise and exactly. It diagnoses support and multiplicity as separate axes rather than inferring one from the other.

7 Pure homology-cover calibrator

Retain the same marked genus-two surface, but now define the distinct tower

$$H_N = \ker\bigl(\Gamma \to H_1(\Sigma; \mathbb{Z}/N\mathbb{Z})\bigr).$$

Its deck group is $(\mathbb{Z}/N\mathbb{Z})^4$. This tower is not residual: its intersection is the commutator subgroup $[\Gamma, \Gamma]$. The owner panel consists of three primitive content-one classes, and the executed schedule is $N = n!$ for $n = 1, \ldots, 8$.

Theorem 7.1 (lift order, count, and period)**.** *For a primitive content-one owner g and every $N \geq 1$:*

(i) the cover degree is N^4;

(ii) the deck image of g has order N;

(iii) the full preimage of the base geodesic has exactly N^3 primitive components; and

(iv) every component has physical period $N\ell(g)$.

Proof. The abelianization of the marked genus-two group is $\mathbb{Z}^4$, and reduction modulo N is onto $(\mathbb{Z}/N\mathbb{Z})^4$, proving the degree. Let $v \in \mathbb{Z}^4$ be the homology vector of g. Since its content is one, Bézout's identity gives an integer linear combination of its coordinates equal to one. Thus $kv = 0$ modulo N forces $N \mid k$, and the order is exactly N.

The lift of the base loop through a deck point follows translation by v. Its orbits have N deck points, so the N^4 deck points split into N^3 cycles. Moreover,

$$H_N \cap \langle g \rangle = \langle g^N \rangle.$$

Therefore a cycle has length $N\ell(g)$. To prove lift primitivity, suppose $g^N = h^q$ with $h \in H_N$ and $q \geq 2$. Centralizers of nontrivial elements in a torsion-free closed surface group are cyclic, so the unique-root property puts $h = g^r$. Then $rq = N$, whereas $h \in H_N$ forces $N \mid r$, which is impossible for $q \geq 2$. □

The theorem also explains why this construction cannot be described as a renormalization of the residual inverse-limit owner without qualification. The tower H_N forgets all information in the commutator subgroup and therefore has nontrivial intersection. Its lifted components are indexed by an abelian deck group, and its clock will be explicitly divided by N. Those are changes to the tower and the time observable, not harmless reformulations. We retain the same base surface only to make the comparison controlled.

Content one is the exact hypothesis behind the clean powers of N. If the homology vector had content $d > 1$, its order modulo N would be $N/\gcd(N, d)$; both component count and lifted period would change with the arithmetic of N. Freezing primitive content-one owners removes that extra variable and makes the quadrant comparison isolate only clock and multiplicity. Primitive homology also certifies that the base geodesic is not a proper power, so the term "primitive component" is earned rather than inferred from a short word.

8 The four renormalization quadrants

Write $x_g = e^{-s\ell(g)}$. Compare the physical clock $N\ell(g)$ with the declared rescaled clock $\ell(g)$, and compare raw multiplicity N^3 with the declared logarithmic geometric mean

$$\log Z^{\mathrm{ren}} = \frac{1}{N^3} \log Z_{\mathrm{panel}}.$$

Theorem 8.1 (four-quadrant law)**.** *Fix $N \geq 1$ and a finite panel of primitive content-one owners. In the coefficientwise topology of the formal power-series ring in independent owner variables, the four registered clock/multiplicity factors are exactly those in Table 2, owner by owner. Among these four quadrants, only the simultaneous $1/N$ clock rescaling and $1/N^3$ logarithmic normalization in Q_{11} recovers every base factor at that level. This statement quantifies over every N and every fixed finite panel; it asserts neither uniformity for a growing panel nor analytic convergence of a global Euler product.*

Table 2: Clock and multiplicity interventions for one owner.

Quadrant	clock	multiplicity	factor and behavior
Q_{00}	physical	raw	$(1-x_g^N)^{-N^3} \to 1$ coefficientwise
Q_{10}	rescaled	raw	$(1-x_g)^{-N^3}$; $[x_g] = N^3$
Q_{01}	physical	geometric mean	$(1-x_g^N)^{-1} \to 1$ coefficientwise
Q_{11}	rescaled	geometric mean	$(1-x_g)^{-1}$ exactly

Proof. Theorem 7.1 supplies N^3 identical lift factors. Physical time gives monomial x_g^N, while the $1/N$ time rescaling gives x_g. Raw multiplicity raises the factor to N^3, and logarithmic normalization divides that exponent by N^3. These choices yield the four displayed factors.

For Q_{00} and Q_{01}, the first nonconstant degree is N, so every fixed prefix is eventually one. For Q_{10}, the coefficient of x_g is N^3 and diverges. The Q_{11} expression is identically the base factor at every level. □

For completeness, the coefficient calculation is exact:

$$[x^d](1-x^m)^{-k} = \begin{cases} 0, & m \nmid d, \\ \binom{k+d/m-1}{d/m}, & m \mid d. \end{cases}$$

The finite replay contains 96 owner/level/quadrant rows and 1,248 integer coefficient rows through degree 12. It validates serialization and all displayed cases, but the theorem follows from the cover calculation for every N.

For a frozen panel P, the argument multiplies ownerwise. In independent variables one obtains

$$Z_{Q_{11},N}(\{x_g\}_{g\in P}) = \prod_{g\in P} \left[(1-x_g)^{-N^3}\right]^{1/N^3} = \prod_{g\in P}(1-x_g)^{-1}.$$

Equivalently, the normalized logarithm is the arithmetic mean of the N^3 identical lift-component logarithms for each owner. No branch of a complex fractional power is needed: the definition is made at the logarithmic formal-series level, where division by N^3 is unambiguous and the constant term is zero.

This panel identity is coefficientwise exact for every N, including $N = 1$, and for each fixed finite panel, but exactness does not make it canonical. The $1/N$ clock rescaling is target-matched to the deck order of every registered owner, and the $1/N^3$ logarithmic normalization is target-matched to the number of lift components. Both quantities are read from the desired base-owner panel in the marked homology cover. The result is therefore a transparent calibration of the cost of making those finite target factors reappear, not a broad theorem-priority claim. An arithmetic construction would still have to derive its interventions intrinsically and control an infinite primitive product.

Proposition 8.2 (uniqueness within scalar clock/exponent renormalizations)**.** *Fix N and one owner. Let $c_N > 0$ multiply the physical lifted period and let $b_N > 0$ multiply the logarithm of the raw N^3-component product. If*

$$\left(1-e^{-sc_N N\ell(g)}\right)^{-b_N N^3} = \left(1-e^{-s\ell(g)}\right)^{-1}$$

for every sufficiently large real s, then

$$c_N = \frac{1}{N}, \qquad b_N = \frac{1}{N^3}.$$

Proof. Put $q = e^{-s\ell(g)}$. Taking logarithms gives

$$b_N N^3 \sum_{r\geq 1} \frac{q^{rc_N N}}{r} = \sum_{r\geq 1} \frac{q^r}{r}.$$

As $q \downarrow 0$, the smallest positive exponent on the left is $c_N N$, while that on the right is one. Equality on an interval forces $c_N N = 1$. The coefficient of the resulting leading term then forces $b_N N^3 = 1$. □

Thus Q_{11} is not merely one successful corner among arbitrary scalar choices: within this registered two-parameter intervention class, its clock and exponent are forced by all-s equality. The proposition does not exclude owner-dependent weights, mixtures between owners, nonlocal transforms, or operator-valued renormalizations. Any such construction would be a further candidate with additional degrees of freedom and would need new controls. The current result keeps the intervention class small enough that the positive equality remains auditable.

The factorial schedule $N = n!$ used by the executable is likewise a finite validation schedule rather than part of Proposition 8.2. The cover theorem and uniqueness argument hold for every positive integer N. Factorials are retained only to compare the calibrator on the same eight-level scale as the closed residual/homology control and to make divisibility between displayed levels explicit.

9 Computational certificates

The residual-tower, closed-control, and homology-calibrator artifacts use deterministic Python-standard-library code. The Round-8 calibrator is frozen by SHA-256

`88d10c3dcdee3387b16414d2c56d4934b6daeef6728acc689855049840850a72`.

Its locked Round-5 owner ledger and validation hashes are

`0c74333b63f6027b16d134f19a320b8148e7fab6f86fa204d213c801106fe825`,
`afdc51ca7ecfbd8777955c7438f08d4580e6b924419a807191e097b0292d9c10`.

Twelve unit tests check the cover formulas, the content-one condition, every quadrant, exact coefficients, owner separation, and Route boundaries. Two isolated builds are byte-identical. The Round-8 core hash is

`a1b588724dacb2ab2986326a7a5e1c6aec654c61538c1465e26564357b568b33`.

The default command `bash experiments/reproduce_round8.sh` verifies the checked-in files without modifying them; `--refresh` is explicit.

The finite replay is an exact executable audit. Its evidence status is

`NUMERICALLY_CERTIFIED`.

The cover and factor theorems have status `PROVED` from the arguments above. These labels are not interchangeable.

The executable certificate mirrors the proof chain. It first validates the immutable owner registry and the content-one homology vectors. It then derives cover degree, deck order, component count, physical and rescaled periods, and the four symbolic factor descriptors. Coefficients are computed as integers from the binomial formula rather than estimated by evaluating a product at floating-point values. Row counts and uniqueness constraints make omissions and duplicates fail closed. Finally, two isolated output trees are compared bytewise and bound to the checked-in hashes.

The earlier residual and closed-control artifacts serve different roles and are not mixed into the positive calibrator. The congruence rows check projective quotient orders and bonding divisibility; the closed rows check the factorial homology lower bound; the Round-8 rows check the four homology-cover quadrants. This separation preserves candidate identity across computational stages. Reproducing all ledgers supports the finite claims and their serialization, while the universal statements in Theorems 3.2, 6.1, 7.1, and 8.1 rest on the written proofs.

10 Adversarial and Route-A analysis

The original residual inverse-limit candidate has weak arithmetic provenance in its congruence specialization but no periodic points. Its formal tuple is

$$(\text{A0\_WEAK\_ARITHMETIC\_RELATION, A1\_FAIL, A2\_FAIL, A3\_FAIL, A4\_FAIL}),$$

with `ROUTE_A_REJECTED`. A1 fails analytically because $\mathrm{Per}(M_\infty) = \varnothing$. A2 is not a failed numerical divisor match; it is not testable on primitive periodic orbits of this owner because there are none.

The pure homology-cover panel is a separate candidate with a new owner declaration, tower, clock, normalization, and fixed-finite-panel scope. For every marked genus-two hyperbolic metric in the stated class, every N, and every fixed finite panel of primitive content-one owners, its exact coefficientwise lift-factor ledger supports `A1_PASS_ANALYTIC`. A0 nevertheless fails: the construction contains no intrinsic rational-prime correspondence or natural $\log p$ scale, and the same cover calculation works throughout that nonarithmetic metric class. This metric-wide validity is the decisive proves-too-much control. A2 fails because no full primitive census or global determinant is defined; A3 and A4 fail because there is no global analytic structure or natural operator lift. Its unchanged tuple is

$$(\text{A0\_FAIL, A1\_PASS\_ANALYTIC, A2\_FAIL, A3\_FAIL, A4\_FAIL}),$$

again `ROUTE_A_REJECTED`. The Q_{11} calibration transfers no credit to the residual candidate, and Route B remains unavailable and unauthorized for both objects.

The four registered quadrants function as comparative calibration controls. Q_{10} shows that support repair alone does not control lift multiplicity, while Q_{01} shows that multiplicity repair alone does not control support. Their failures and the positive Q_{11} identity are exact within the same fixed finite panel and scalar clock/exponent intervention frame. Because Q_{11} changes the owner declaration, tower, clock, normalization, and panel scope, it cannot rescue the residual candidate or inherit arithmetic Route credit for it.

11 Limitations and open obligations

The no-period theorem assumes a normal residual tower and one common clock. It does not cover arbitrary nonnormal chains without additional conjugator control. The three cusped diagnostic rows lack full conjugacy-primitivity certificates, so their computed times are loop-closing diagnostics rather than asserted primitive minimal periods. The closed control avoids that issue through primitive homology, but its full residual-core quotient orders are not enumerated.

The renormalized identity concerns a fixed finite owner panel. The pure homology tower is not residual, and no interchange between a tower limit and an infinite product over all primitive geodesics is justified. Growing panels could introduce uncontrolled orbit proliferation and would require a new convergence theorem. The logarithmic $1/N^3$ normalization is explicitly imposed from deck multiplicity; it is not derived from arithmetic weights. Finally, no target divisor, functional equation, transfer operator, or quantum lift appears. These limitations block any promotion from a finite calibration identity to a Hilbert–Pólya claim.

There is no contradiction between the negative residual-tower theorem and the positive homology-cover identity. The former fixes a coherent inverse-limit point, a residual tower, and the physical clock;

the latter fixes a base-owner panel, replaces the tower by a nonresidual abelian cover, rescales time at each level, and averages the logarithms of distinct lift components. Since both the quantified objects and the observables change, Q_{11} does not restore periodic points to M_∞. It constructs a separate finite factor identity.

A natural next question is whether a canonically weighted growing panel admits a locally uniform logarithmic limit after both interventions. Answering it would require at least a primitive-owner counting law, control of lift multiplicities uniformly over that population, and bounds that justify summing before taking the tower limit. None of those inputs is contained in the 96-row calibrator. Coefficientwise equality for each fixed finite panel does not supply uniformity as the panel grows.

12 Conclusion

This comparative methods and calibration note keeps two candidates separate. For the residual candidate, the prior general aperiodicity mechanism specializes to the literal factorial congruence chain: the inverse-limit flow has no periodic point, and every fixed finite same-owner panel escapes every fixed physical-time or coefficient window. For the distinct homology calibrator, the four-quadrant calculation shows that, within the registered scalar clock/exponent intervention class, the target-matched Q_{11} combination recovers each fixed finite-panel factor coefficientwise for every N; neither intervention alone does. The exact specialization, owner audit, and calibration are the manuscript's bounded contributions. They do not supply a broad theorem-priority claim, a canonical renormalization, or arithmetic specificity. The residual tuple remains (A0_WEAK_ARITHMETIC_RELATION, A1_FAIL, A2_FAIL, A3_FAIL, A4_FAIL) and the homology tuple remains (A0_FAIL, A1_PASS_ANALYTIC, A2_FAIL, A3_FAIL, A4_FAIL); both remain `ROUTE_A_REJECTED`, Q_{11} transfers no credit between them, and Route B remains unavailable.

Data and Code Availability

All data are repository artifacts under `papers/27-congruence-inverse-limit-no-go/results`. Deterministic source code, tests, freeze contracts, validation reports, and verify-default reproduction scripts are available in the corresponding `code` and `experiments` directories. No proprietary, personal, prime-target, or Riemann-zero dataset was used.

Ethics Declaration

This theoretical and computational mathematics study involved no human participants, personal data, animals, clinical material, or field intervention. Institutional ethics approval and informed consent were not applicable.

Author Contributions

Liang Wang: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing–Original Draft, Writing–Review & Editing, Visualization, and Project Administration.

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